STATISTICS 5.3.

Probability

Random events happen by chance. → fair, unbiased

Probability is a measure of how likely they are.

1 The basics

- outcome / elementary event = the result of an experiment → each repeat = a 'trial'
- events = 'groups' of ≥ 1 outcomes
- when all outcomes are equally likely (equiprobable)

  → calculate probability :

$$P(\text{event}) = \frac{\text{No. of outcomes when event happens}}{\text{Total no. of possible outcomes}}$$

P

0 ⟷ 1

impossible — certain

EXAMPLE Suppose I've got a bag with 15 balls in — 5 red, 6 blue and 4 green.

If I take a ball out without looking, then any ball is equally likely — there are 15 possible outcomes. Of these 15 outcomes, 5 are red, 6 are blue and 4 are green. And so...

$P(\text{red ball}) = \frac{5}{15} = \frac{1}{3}$ $P(\text{blue ball}) = \frac{6}{15} = \frac{2}{5}$ $P(\text{green ball}) = \frac{4}{15}$

You can find the probability of either a red or a green ball in a similar way...

$P(\text{red or green ball}) = \frac{9}{15} = \frac{3}{5}$

- exhaustive events : contains all possible outcomes of an experiment

  e.g. : $P(\text{red}) + P(\text{not red}) = 1$
- expectation: estimate likelihood of event in a series of trials based on probability

2 Mutually exclusive events, the addition law

↳ cannot both occur at the same time.

e.g.: Roll a die once. I can get a 1 OR a 5. CAN'T get a 1 AND a 5.

$P(1 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$

$P(A) + P(A') = 1$

$\boxed{P(A \text{ or } B) = P(A) + P(B)}$

↕

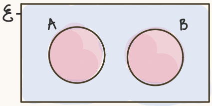
$\boxed{P(A \cup B) = P(A) + P(B)}$

* Venn diagrams

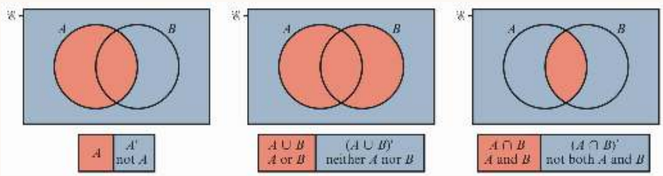
if mutually exclusive →

· no overlap · $P(A \cap B) = 0$

$A \cap B = \emptyset$ ← empty set



then:



A and B overlaps

= has common favorable outcome

= NOT mutually exclusive

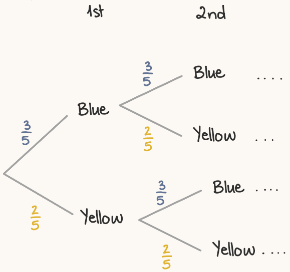
3. Independent events; the multiplication law

↳ occur with NO effect on each other → the other can occur or not occur, doesn't matter.

e.g.: A bag has 3 blue marbles and 2 yellow marbles.
Pick 1, record color. REPLACE it. Pick another.
↳ ensures there are always 5 marbles to pick from.

$\boxed{P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)}$

* Tree diagrams



$P(BB) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

$P(BY) = \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$

$P(YB) = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$

$P(YY) = \frac{2}{5} \times \frac{2}{5} = \frac{4}{5}$

4. Conditional probability

A child is selected at random from a group of 11 boys and nine girls, and one of the girls is called Rose. Find the probability that Rose is selected, given that a girl is selected.

Answer

P(Rose is selected | a girl is selected) $= \frac{1}{9}$

The additional information, 'given that a girl is selected', reduces the number of possible selections from 20 to 9, and Rose is one of those nine girls.

probability of A, given that B has already happened

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

5. Dependent events and conditional probability

↳ occurence of 1 affects the other

e.g.: making a selection without replacement

→ 2nd outcome depends on 1^{st} outcome

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A)$$
$$P(B \text{ and } A) = P(B \cap A) = P(B) \times P(A|B)$$

$$P(A) \times P(B|A) = P(B) \times P(A|B)$$

* if B is independent of A:

$$P(B|A) = P(B|A') = P(B)$$

probability that B occurs, given A occurs = probability that B occurs given A doesn't occur

↳ A and B are not affected by each other